Comprehensive UAT Testing Guide

GCC Database Solution

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Project: GCC Web Application (ASP.NET Core 8.0)
Environment: UAT / System Test

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1. Overview & Purpose

This guide provides a **complete, end-to-end** User Acceptance Testing plan for the GCC Database Solution. It covers:

- **All 6 user roles**: Public, External Client, Partner, Analyst, Admin, API Integration
- **All application pages**: 30+ pages across authentication, certificates, admin, and API
- **14-state certificate workflow**: Every status transition with role-based access
- **Permission verification**: What each role can and cannot do
- **Requirements validation**: Mapping test cases back to documented requirements

How to Use This Guide

1. **Before starting**: Set up test accounts per Section 2
2. **Execute by module**: Work through Modules 1-10 in order
3. **Record results**: Use Pass/Fail/Blocked columns in each test table
4. **Log defects**: Use the template in Section 25
5. **Track progress**: Update the summary in Section 26

2. Test Credentials & Setup

Test Accounts

Username	Password	Role	Company	Purpose
admin	TestPass123!	Administrator	N/A	Full system access
analyst	TestPass123!	Analyst	N/A	Certificate management
external	TestPass123!	External Client	(Assigned Company)	Company-scoped view
(no account)	N/A	Public	N/A	Unauthenticated access

Environment URLs

Environment	URL
Local Dev	http://localhost:5101
UAT	(TBD - configure per deployment)

Pre-Test Checklist

- [ ] Application is running and accessible
- [ ] **Run `UAT_TEST_DATA_SETUP.sql`** against UAT database
- [ ] Database seeded with test data (4 companies, 6 products, 8 certificates, 7 tags)
- [ ] 8 certificates in various statuses (Active x3, Draft, Archived, Expired, PendingReview, Active-expiring)
- [ ] 4 companies with products (EcoProduct Co, GreenTech Industries, TestCo UAT, OtherCo)
- [ ] External user associated with TestCo UAT (Company Id 3)
- [ ] PDF test file available (< 10MB) for upload testing
- [ ] CSV test files available for import testing
- [ ] Browser DevTools available for API/network inspection
- [ ] For accessibility testing: Screen reader installed (NVDA/JAWS/VoiceOver)

3. Test Case Format & Status Key

Each test case includes: **ID**, **Description**, **Pre-conditions**, **Steps**, **Expected Result**, and **Status**.

Status	Icon	Meaning
Pass	✓	Test passed as expected
Fail	✗	Test did not produce expected result
Blocked	⊘	Cannot execute due to dependency
Not Run	☐	Not yet executed

4. Module 1: Authentication & Account Management

1.1 Login Flow

ID	Description	Steps	Expected Result	Status
TC-AUTH-001	Valid admin login	1. Navigate to /Account/Login 2. Enter admin/TestPass123! 3. Click Sign In	Redirect to Dashboard or Internal Tile View	
TC-AUTH-002	Valid analyst login	1. Navigate to /Account/Login 2. Enter analyst/TestPass123! 3. Click Sign In	Redirect to Internal Tile View	
TC-AUTH-003	Valid external login	1. Navigate to /Account/Login 2. Enter external/TestPass123! 3. Click Sign In	Redirect to External Tile View	
TC-AUTH-004	Invalid password	1. Navigate to /Account/Login 2. Enter admin/WrongPassword 3. Click Sign In	Error message displayed, no redirect	
TC-AUTH-005	Invalid username	1. Navigate to /Account/Login 2. Enter nonexistent/TestPass123! 3. Click Sign In	Error message displayed	
TC-AUTH-006	Empty fields submission	1. Navigate to /Account/Login 2. Leave both fields empty 3. Click Sign In	Validation errors shown for required fields	
TC-AUTH-007	Account lockout	1. Enter wrong password 5+ times for same account	Account locked message displayed	
TC-AUTH-008	Remember Me functionality	1. Login with Remember Me checked 2. Close browser 3. Reopen and navigate to app	User remains authenticated	

1.2 Two-Factor Authentication (2FA)

ID	Description	Steps	Expected Result	Status
TC-AUTH-009	Enable 2FA	1. Login as admin 2. Navigate to Account > Two Factor Preferences 3. Enable 2FA 4. Scan QR code 5. Enter TOTP code	2FA enabled, recovery codes displayed	
TC-AUTH-010	Login with 2FA enabled	1. Login with username/password 2. Enter TOTP code from authenticator	Successful login after 2FA verification	
TC-AUTH-011	Invalid 2FA code	1. Login with username/password 2. Enter wrong TOTP code	Error message, not authenticated	
TC-AUTH-012	Disable 2FA (Admin)	1. Login as admin 2. Navigate to User Management 3. Select user with 2FA 4. Click Disable 2FA	2FA disabled for target user	

1.3 Password Management

ID	Description	Steps	Expected Result	Status
TC-AUTH-013	Forgot password flow	1. Click Forgot Password 2. Enter email 3. Check email for reset link 4. Set new password	Password reset successful, can login with new password	
TC-AUTH-014	Force password reset	1. Admin flags user for password reset 2. User logs in 3. User is redirected to password reset	User must set new password before accessing system	

1.4 Session & Authorization

ID	Description	Steps	Expected Result	Status
TC-AUTH-015	Session timeout	1. Login 2. Leave idle beyond session timeout 3. Attempt action	Redirect to login page	
TC-AUTH-016	Access denied - External to Admin	1. Login as external 2. Navigate directly to /Admin/UserManagement	Access Denied page or redirect	
TC-AUTH-017	Access denied - Analyst to Admin	1. Login as analyst 2. Navigate directly to /Admin/UserManagement	Access Denied page or redirect	
TC-AUTH-018	Logout	1. Login as any user 2. Click Logout	Redirected to login page, session cleared	

5. Module 2: Public User Flows

Role: Unauthenticated user (no login required)

2.1 Public Certificate Tile View

ID	Description	Steps	Expected Result	Status
TC-PUB-001	Access public tile view	1. Navigate to /Certificate/CertificateTileView (no login)	Page loads with certificate tiles visible	
TC-PUB-002	Infinite scroll loading	1. View tile view 2. Scroll to bottom	Additional certificates load automatically	
TC-PUB-003	Search by certificate number	1. Type certificate number in search bar	Autocomplete suggestions appear, results filter	
TC-PUB-004	Search by company name	1. Type company name in search bar	Autocomplete suggestions with categorized results	
TC-PUB-005	Filter by certification type	1. Open filter panel 2. Select a certification type checkbox	Tiles filter to show only selected type	
TC-PUB-006	Filter by status	1. Open filter panel 2. Select "Active" status	Only Active certificates shown	
TC-PUB-007	Multiple filters combined	1. Select certification type filter 2. Select status filter 3. Enter search text	Results match ALL applied filters	
TC-PUB-008	Clear all filters	1. Apply multiple filters 2. Click clear/reset	All certificates shown, filters cleared	
TC-PUB-009	Shareable URL	1. Apply filters 2. Copy URL 3. Open in new incognito window	Same filters applied, same results shown	
TC-PUB-010	No edit/create buttons visible	1. View tile view as public user	No Add, Edit, or Delete buttons anywhere	
TC-PUB-011	No internal tags visible	1. View certificate tiles	Internal tag column/data is NOT shown	
TC-PUB-012	No navigation menu	1. View page as unauthenticated user	Only public tile view accessible, no admin/management nav links	
TC-PUB-013a	Draft certificates NOT visible	1. Ensure a Draft certificate exists 2. View public tile view	Certificate NOT shown (only Active visible to public)	

ID	Description	Steps	Expected Result	Status
TC-PUB-013b	Archived certificates NOT visible	1. Ensure an Archived certificate exists 2. View public tile view	Certificate NOT shown	
TC-PUB-013c	Filter options from database	1. Open filter panel on public tile view	Filter options dynamically loaded from database, not hardcoded	

2.2 Public Certificate Details

ID	Description	Steps	Expected Result	Status
TC-PUB-013	View single certificate	1. Click on a certificate tile	Certificate detail page loads with: Certificate Number, Company, Product, Status, Dates, Certification Type	
TC-PUB-014	Certificate preview image	1. View certificate details	PDF thumbnail preview shown (or placeholder)	
TC-PUB-015	Download certificate PDF	1. View certificate with PDF uploaded 2. Click Download	PDF file downloads	
TC-PUB-016	No edit buttons on details	1. View certificate detail page as public	No Edit or Delete buttons visible	
TC-PUB-017	Certificate number in URL	1. View certificate details	URL uses certificate number (e.g., /Certificate/CertificateSingleView/24-0366), NOT database ID	

6. Module 3: External User Flows

Role: External Client (login: external / TestPass123!)
Key Rule: Can ONLY see data for their associated company/partner

3.1 External Certificate Tile View

ID	Description	Steps	Expected Result	Status
TC-EXT-001	Access external tile view	1. Login as external 2. Verify redirect to /Certificate/CertificateTileViewExternal	External tile view loads	
TC-EXT-002	Only own company data visible	1. View tile view 2. Check all visible certificates	ALL certificates belong to external user's associated company only	
TC-EXT-003	Cannot see other company data	1. Search for a certificate from a different company	No results returned for other companies	
TC-EXT-004	No Add Certificate button	1. View external tile view	No "Add" or "Create" button visible	
TC-EXT-005	No Edit buttons	1. View tiles	No Edit button/icon on any certificate	

ID	Description	Steps	Expected Result	Status
TC-EXT-006	No Delete buttons	1. View tiles	No Delete button/icon on any certificate	
TC-EXT-007	No internal tags visible	1. View certificate tiles	Internal Tag column is NOT displayed	
TC-EXT-008	External tags visible	1. View certificate with external tags	External tags ARE displayed	
TC-EXT-009	Search within own data	1. Use search bar 2. Enter text	Search works but only within company's certificates	
TC-EXT-010	Filter within own data	1. Apply certification type filter	Filters work within company's data only	
TC-EXT-011	Infinite scroll	1. Scroll to bottom	More certificates from own company load	
TC-EXT-012	Bulk ZIP download	1. Select multiple certificates 2. Click download	ZIP file downloads with selected certificate PDFs	

3.2 External Certificate Details

ID	Description	Steps	Expected Result	Status
TC-EXT-013	View own certificate detail	1. Click on a certificate tile	Detail view shows certificate data (read-only)	
TC-EXT-014	No edit/delete on details	1. View certificate detail page	No Edit or Delete buttons	
TC-EXT-015	Cannot access other company cert via URL	1. Manually enter URL for a certificate from another company	Access denied or certificate not found	

3.3 External Product Management

ID	Description	Steps	Expected Result	Status
TC-EXT-016	Access product management	1. Navigate to /Product/Management	Page loads with products	
TC-EXT-017	Only own company products	1. View product list	Only products for external user's company shown	
TC-EXT-018	No Actions column	1. View product table	Actions column (Edit/Delete) is NOT visible	
TC-EXT-019	No Add Product button	1. View product page	No Add/Create button visible	
TC-EXT-020	Product data is read-only	1. Try to interact with product data	No inline editing available	

3.4 External Navigation Verification

ID	Description	Steps	Expected Result	Status
TC-EXT-021	Navigation menu items	1. Check main navigation menu	Only see: Certificates, Products (NO Admin, NO Companies, NO Dashboard)	
TC-EXT-022	Cannot access admin pages	1. Navigate directly to /Admin/UserManagement	Access denied	
TC-EXT-023	Cannot access certificate create	1. Navigate directly to /Certificate/Create	Access denied	
TC-EXT-024	Cannot access company management	1. Navigate directly to /Company/Index	Access denied	

7. Module 4: Analyst User Flows

Role: Analyst (login: analyst / TestPass123!)
Key Rule: Can create/edit certificates, manage workflow, but NO admin or delete access

4.1 Internal Certificate Tile View

ID	Description	Steps	Expected Result	Status
TC-ANL-001	Access internal tile view	1. Login as analyst	Internal tile view loads at /Certificate/CertificateTileViewInternal	
TC-ANL-002	See all certificates	1. View tile view	Certificates from ALL companies visible (not scoped)	
TC-ANL-003	Internal tags visible	1. View tiles	Internal tags column IS visible	
TC-ANL-004	Add Certificate button visible	1. View tile view	Add/Create New button IS visible	
TC-ANL-005	Edit button visible	1. View certificate tile	Edit button/icon IS visible on each tile	
TC-ANL-006	Search across all data	1. Search for certificate	Results span all companies and types	
TC-ANL-007	All filters available	1. Open filter panel	Can filter by: Certification Type, Status, Company, Partner, Tags	
TC-ANL-008	Shareable URL with filters	1. Apply filters 2. Copy URL 3. Share with another analyst	Same filters/results when opened	
TC-ANL-009	Bulk ZIP download	1. Select multiple certificates 2. Download	ZIP file with PDFs downloads	

4.2 Certificate Create Flow

ID	Description	Steps	Expected Result	Status
TC-ANL-010	Navigate to create page	1. Click Add/Create New from tile view	/Certificate/Create page loads	

ID	Description	Steps	Expected Result	Status
TC-ANL-011	Auto-generated certificate number	1. View create form	Certificate Number field pre-populated with next sequential number (YY-NNNN format)	
TC-ANL-012	Required field validation	1. Leave all fields empty 2. Click Save	Validation errors shown for required fields	
TC-ANL-013	Select company	1. Select a company from dropdown	Company selected, associated products may update	
TC-ANL-014	Select product	1. Select a product from dropdown	Product selected	
TC-ANL-015	Select certification type	1. Select parent certification type	Subcategory dropdown populates with child types	
TC-ANL-016	Select subcategory	1. Select subcategory from populated dropdown	Subcategory selected, custom fields may load	
TC-ANL-017	Custom fields load dynamically	1. Select a certification type that has custom fields	Custom field inputs appear below cert type selection	
TC-ANL-018	Set effective date	1. Enter or pick effective date	Date accepted in correct format	
TC-ANL-019	Set expiration date	1. Enter expiration date (after effective date)	Date accepted	
TC-ANL-020	Expiration before effective date	1. Set expiration date BEFORE effective date	Validation error shown	
TC-ANL-021	Upload PDF certificate	1. Click upload zone or drag PDF file 2. Verify preview	File name shown, preview generated, success toast	
TC-ANL-022	Upload invalid file type	1. Try to upload a .docx or .jpg file	Error: Invalid file type	
TC-ANL-023	Upload oversized file	1. Try to upload PDF > 10MB	Error: File too large	
TC-ANL-024	Add external tags	1. Select external tags from dropdown	Tag badges appear, can be removed with X	
TC-ANL-025	Add internal tags	1. Select internal tags from dropdown	Tag badges appear	

ID	Description	Steps	Expected Result	Status
TC-ANL-026	Add product identifiers	1. Click Add Product ID 2. Fill in Type (ASIN), Value, Status 3. Save	Product ID appears in table below form	
TC-ANL-027	Add multiple product IDs	1. Add 3+ product IDs of different types (ASIN, UPC, GTIN)	All appear in product ID table	
TC-ANL-028	Edit product identifier	1. Click edit on a product ID row 2. Modify value 3. Save	Updated value shown in table	
TC-ANL-029	Delete product identifier	1. Click delete on a product ID row 2. Confirm	Product ID removed from table	
TC-ANL-030	Import product IDs via CSV	1. Click Import on product ID section 2. Upload CSV	Product IDs populated from CSV	
TC-ANL-031	Enter Certification Results	1. Enter text in Certification Results field (optional)	Text accepted (up to 1000 chars)	
TC-ANL-031a	Cert Results visible in tile view	1. Save certificate with Cert Results 2. Return to internal tile view	Certification Results column visible with entered value	
TC-ANL-031b	Cert Results in CSV export	1. Export certificates from tile view 2. Open CSV	Certification Results column present in exported CSV	
TC-ANL-031c	Pathway dropdown on create form	1. Select Certification Type that has pathways configured	Pathway dropdown appears with options scoped to selected cert type	
TC-ANL-031d	Pathway hidden when no pathways	1. Select Certification Type with NO pathways configured	Pathway dropdown is hidden (not shown)	
TC-ANL-031e	Pathway selection saved	1. Select Pathway 2. Save certificate 3. Re-open edit	Selected Pathway value persists on certificate	
TC-ANL-031f	Soft warning - Amazon cert missing Cert Results	1. Select Partner = "Amazon" 2. Leave Cert Results empty 3. Save	Warning message displayed (NOT blocking). Certificate saves successfully.	**Pending Q4**
TC-ANL-032	Set partner	1. Select partner (Amazon/Novi/Other/None)	Partner set	
TC-ANL-033	Set location and country	1. Enter location and country	Fields populated	
TC-ANL-034	Analyst available statuses	1. Check Status dropdown options	Should see 12 of 14 statuses (all except Rejected and Revoked)	

ID	Description	Steps	Expected Result	Status
TC-ANL-035	Save certificate	1. Fill all required fields 2. Click Save	Certificate created, redirect to details, success notification	
TC-ANL-036	Save & Add Another	1. Fill all required fields 2. Click Save & Add Another	Certificate created, redirect back to blank Create form	
TC-ANL-037	Unsaved changes warning	1. Fill some fields 2. Try to navigate away without saving	Warning prompt about unsaved changes	

4.3 Certificate Edit Flow

ID	Description	Steps	Expected Result	Status
TC-ANL-038	Open edit page	1. From tile view, click Edit on a certificate	Edit page loads with all current data pre-populated	
TC-ANL-039	All fields pre-populated	1. Verify all fields on edit page	Company, Product, Type, Subcategory, Dates, Tags, Product IDs, Custom Fields, Cert Results, Pathway all pre-filled	
TC-ANL-040	Edit company	1. Change company dropdown	Company updated	
TC-ANL-041	Edit certification type	1. Change certification type	Subcategory and custom fields update	
TC-ANL-042	Add new tag on edit	1. Add a tag that wasn't previously assigned	Tag badge appears	
TC-ANL-043	Remove existing tag	1. Click X on an existing tag badge	Tag removed	
TC-ANL-044	Replace PDF	1. Upload new PDF on edit page	New file shown, old replaced	
TC-ANL-045	Change status on edit	1. Change status dropdown	Status updated (within allowed transitions)	
TC-ANL-046	Save edited certificate	1. Make changes 2. Click Save Changes	Changes saved, redirect to details, success notification	
TC-ANL-047	Certificate number unchanged	1. Open edit 2. Check certificate number field	Certificate number cannot be changed on edit	

4.4 Certificate Details View (Internal)

ID	Description	Steps	Expected Result	Status
TC-ANL-048	View certificate details	1. Click on certificate from tile view	Detail page shows ALL fields including internal tags	
TC-ANL-049	PDF preview and download	1. View details of cert with PDF	Thumbnail preview visible, Download button works	
TC-ANL-050	Click-to-zoom preview	1. Click on thumbnail preview	Full-size modal opens with certificate image	
TC-ANL-051	Edit button on details	1. View details page	Edit button visible and functional	

ID	Description	Steps	Expected Result	Status
TC-ANL-052	Certificate number displayed (not ID)	1. View details page	Page shows Certificate Number (e.g., 24-0366), NOT database ID	

4.5 Analyst Navigation & Restrictions

ID	Description	Steps	Expected Result	Status
TC-ANL-053	Navigation menu	1. Check navigation	See: Certificates, Dashboard, Companies & Products (NOT Administration)	
TC-ANL-054	Cannot delete certificates	1. Check tile view and details	No Delete button/option visible for analyst	
TC-ANL-055	Cannot access admin pages	1. Navigate to /Admin/UserManagement directly	Access denied	
TC-ANL-056	Can access Dashboard	1. Click Dashboard in nav	Dashboard page loads with statistics	
TC-ANL-057	Can access Companies	1. Navigate to /Company/Index	Company list loads	
TC-ANL-058	Can manage products	1. Navigate to /Product/Management	Product page with CRUD operations	

4.6 Certificate Import Access (Analyst - DENIED)

> **Note**: Certificate Import is Admin-only. Analysts CANNOT access the bulk import page.

ID	Description	Steps	Expected Result	Status
TC-ANL-059	Analyst cannot access certificate import	1. Login as analyst 2. Navigate to /CertificateImport/Index	Access denied / redirect	
TC-ANL-060	No import menu item for analyst	1. Check navigation menu as analyst	Certificate Import link NOT visible in nav	

4.7 Additional Analyst Validations

ID	Description	Steps	Expected Result	Status
TC-ANL-061	Cascading company-product dropdown	1. On Create form, select a Company 2. Open Product dropdown	Product dropdown filters to show only products linked to selected company	
TC-ANL-062	Company change resets product	1. Select Company A and Product A-1 2. Change to Company B	Product dropdown resets/filters to Company B products only	
TC-ANL-063	Database-driven filter options	1. Open filter panel on internal tile view	All filter options loaded from database, not hardcoded	

8. Module 5: Admin User Flows

Role: Administrator (login: admin / TestPass123!)
Key Rule: Full system access including all management, delete, and configuration

5.1 Admin Dashboard

ID	Description	Steps	Expected Result	Status
TC-ADM-001	Access admin dashboard	1. Login as admin	Dashboard loads with system statistics	
TC-ADM-002	Statistics cards display	1. View dashboard	Cards show: Total Certificates, Active, Expired, Companies, Users, etc.	
TC-ADM-003	Stat card downloads	1. Click download on a stat card	Data downloads as CSV	

5.2 User Management

ID	Description	Steps	Expected Result	Status
TC-ADM-004	Access user management	1. Navigate to Admin > User Management	User list table loads	
TC-ADM-005	Search users	1. Type in search box	Table filters in real-time	
TC-ADM-006	Filter by role	1. Select role from filter dropdown	Only users with selected role shown	
TC-ADM-007	Filter by status	1. Select Active/Inactive filter	Users filtered by status	
TC-ADM-008	Filter by company	1. Select company from filter	Users filtered by company association	
TC-ADM-009	Create new user	1. Click Add New 2. Fill: Username, Email, Password, Role, Company 3. Save	User created, appears in table, success toast	
TC-ADM-010	Create user - duplicate email	1. Try to create user with existing email	Error: Email must be unique	
TC-ADM-011	Create user - duplicate username	1. Try to create user with existing username	Error: Username must be unique	
TC-ADM-012	Edit existing user	1. Click Edit on a user 2. Change role 3. Save	User updated, success toast	
TC-ADM-013	Delete user	1. Click Delete on a user 2. Confirm in modal	Confirmation modal appears FIRST, then user deleted, success toast	
TC-ADM-014	Cannot delete admin user	1. Try to delete the admin user	Operation prevented, error message	
TC-ADM-015	Reset user 2FA	1. Click Reset 2FA on user with 2FA enabled	2FA reset for user	
TC-ADM-016	Export users CSV	1. Click Export	CSV file downloads with all user data	
TC-ADM-017	Download user import template	1. Click Template	CSV template downloads with correct headers	
TC-ADM-018	Import users from CSV	1. Click Import 2. Upload valid CSV	Users imported, success notification	

5.3 Role Management

ID	Description	Steps	Expected Result	Status
TC-ADM-019	Access role management	1. Navigate to Admin > Role Management	Role list with permission matrix loads	
TC-ADM-020	View role permissions	1. Click on a role	Permission checkboxes displayed by category (6 categories, 24 permissions)	
TC-ADM-021	Create custom role	1. Click Add Role 2. Enter Name/Description 3. Select permissions 4. Save	Role created with selected permissions	
TC-ADM-022	Edit role permissions	1. Click Edit on role 2. Change permissions 3. Save	Permissions updated	
TC-ADM-023	Delete custom role	1. Click Delete on custom role 2. Confirm	Role deleted (confirmation modal before, toast after)	
TC-ADM-024	Cannot delete Admin role	1. Try to delete Admin role	Operation prevented	
TC-ADM-025	Select All permissions	1. Edit role 2. Click Select All	All permissions checked	
TC-ADM-026	Clear All permissions	1. Edit role 2. Click Clear All	All permissions unchecked	
TC-ADM-027	Export roles CSV	1. Click Export	Roles CSV downloads	

5.4 Certificate Type Management

ID	Description	Steps	Expected Result	Status
TC-ADM-028	Access certificate types	1. Navigate to Admin > Data Management > Certificate Types	Certificate type hierarchy loads	
TC-ADM-029	View parent/child hierarchy	1. View certificate types list	Parent types shown with associated subcategories	
TC-ADM-030	Create parent cert type	1. Click Add 2. Enter Name, Description 3. Save	New parent certification type created	
TC-ADM-031	Create subcategory	1. Click Add subcategory under a parent 2. Enter Name 3. Save	Subcategory created under parent	
TC-ADM-032	Edit cert type (Amazon Code)	1. Edit a cert type 2. Set AmazonCode field 3. Save	Amazon Code saved for API mapping	
TC-ADM-033	Delete cert type	1. Delete a cert type with no certificates 2. Confirm	Type deleted	
TC-ADM-034	Help modal	1. Click Help button	Help modal displays hierarchy explanation	

5.5 Certificate Field Management

ID	Description	Steps	Expected Result	Status
TC-ADM-035	Access certificate fields	1. Navigate to Admin > Certificate Fields	Custom fields list loads	

ID	Description	Steps	Expected Result	Status
TC-ADM-036	Create text field	1. Add new field: Type=Text, Name, Required/Optional 2. Save	Text field created	
TC-ADM-037	Create number field	1. Add new field: Type=Number 2. Save	Number field created	
TC-ADM-038	Create date field	1. Add new field: Type=Date 2. Save	Date field created	
TC-ADM-039	Create dropdown field	1. Add new field: Type=Select, add options 2. Save	Dropdown field created with options	
TC-ADM-040	Associate field with cert type	1. Set certification type association on field	Field appears when that cert type is selected on Create/Edit	
TC-ADM-041	Edit custom field	1. Edit field name/type 2. Save	Field updated	
TC-ADM-042	Delete custom field	1. Delete field 2. Confirm	Field deleted (confirmation modal, then toast)	

5.6 Tag Management

ID	Description	Steps	Expected Result	Status
TC-ADM-043	Access tag management	1. Navigate to Admin > Tags	Tag list loads	
TC-ADM-044	Filter by tag type	1. Filter by Internal/External	Tags filtered by type	
TC-ADM-045	Create external tag	1. Add new tag: Name, Type=External, Active 2. Save	External tag created (visible to external users)	
TC-ADM-046	Create internal tag	1. Add new tag: Name, Type=Internal, Active 2. Save	Internal tag created (NOT visible to external users)	
TC-ADM-047	Edit tag	1. Edit tag name/status 2. Save	Tag updated	
TC-ADM-048	Delete tag	1. Delete tag 2. Confirm	Tag deleted (confirmation modal, then toast)	
TC-ADM-049	Export tags CSV	1. Click Export	Tags CSV downloads	
TC-ADM-050	Download tag template	1. Click Template	Import template CSV downloads	
TC-ADM-051	Import tags CSV	1. Click Import 2. Upload CSV	Tags imported	

5.7 Partner Management

ID	Description	Steps	Expected Result	Status
TC-ADM-052	Access partners	1. Navigate to Admin > Partners	Partner list loads	
TC-ADM-053	Core partners exist	1. View list	Amazon, Novi, Other partners present	

ID	Description	Steps	Expected Result	Status
TC-ADM-054	Create partner	1. Click Add 2. Enter partner details 3. Save	Partner created	
TC-ADM-055	Edit partner	1. Edit partner details 2. Save	Partner updated	
TC-ADM-056	Delete partner	1. Delete partner 2. Confirm	Partner deleted	
TC-ADM-057	Export partners CSV	1. Click Export	Partners CSV downloads	
TC-ADM-058	No Template/Import buttons	1. View Partners page	NO Template button, NO Import button (Export + Add only)	

5.8 Product Type Management

ID	Description	Steps	Expected Result	Status
TC-ADM-059	Access product types	1. Navigate to Admin > Product Types	Product type list loads	
TC-ADM-060	Create product type	1. Add new product type 2. Save	Product type created	
TC-ADM-061	Edit product type	1. Edit product type 2. Save	Product type updated	
TC-ADM-062	Delete product type	1. Delete product type 2. Confirm	Product type deleted	

5.9 CSF Attributes

ID	Description	Steps	Expected Result	Status
TC-ADM-063	Access CSF attributes	1. Navigate to Admin > CSF Attributes	CSF attribute list loads	
TC-ADM-064	CRUD operations	1. Create, Read, Update, Delete CSF attributes	All CRUD operations work	

5.10 Pathway Management

ID	Description	Steps	Expected Result	Status
TC-ADM-065	Access pathways	1. Navigate to Admin > Pathways	Pathway list loads	
TC-ADM-066	Create pathway	1. Add new pathway with cert type association 2. Save	Pathway created	
TC-ADM-067	Edit pathway	1. Edit pathway 2. Save	Pathway updated	
TC-ADM-068	Delete pathway	1. Delete pathway 2. Confirm	Pathway deleted	
TC-ADM-068a	Pathway scoped to cert type	1. Create pathway with Cert Type = "Energy Savings" 2. Go to Create Certificate 3. Select type	Pathway appears in dropdown for matching type only	Deferred
TC-ADM-068b	General pathway visible for all types	1. Create pathway with no cert type scoping 2. Create cert with any type	General pathway appears regardless of cert type	Deferred
TC-ADM-068c	Pathway visible to external users	1. Set pathway on certificate 2. Login as external	Verify if pathway is shown on external detail view	Pending

5.11 Partner API Management

ID	Description	Steps	Expected Result	Status
TC-ADM-069	Access Partner API page	1. Navigate to Admin > API Management > Partner API	API key management loads	
TC-ADM-070	Generate API key	1. Generate new API key for partner	Key generated, displayed	
TC-ADM-071	View API usage	1. Check usage statistics for partner	Usage stats displayed	
TC-ADM-072	Revoke API key	1. Revoke existing API key	Key revoked, no longer functional	

5.12 Google Analytics Configuration

ID	Description	Steps	Expected Result	Status
TC-ADM-073	Access GA settings	1. Navigate to Admin > Configuration > Google Analytics	GA settings page loads	
TC-ADM-074	Enable/Disable tracking	1. Toggle GA tracking on/off 2. Save	Setting persisted	

5.13 Certificate Import (Admin Only)

> **Note**: Certificate Import requires AdminOnly policy. Analysts and External users cannot access this page.

ID	Description	Steps	Expected Result	Status
TC-ADM-075	Access certificate import	1. Login as admin 2. Navigate to /CertificateImport/Index	Import page loads	
TC-ADM-076	Download import template	1. Click Template button	CSV template with correct headers including: CERTIFICATE_NUMBER, COMPANY_NAME, PRODUCT_NAME, CERTIFICATION_TYPE, SUBCATEGORY, EFFECTIVE_DATE, EXPIRATION_DATE, STATUS, CERTIFICATION_RESULTS, PATHWAY, PARTNER, LOCATION, COUNTRY	
TC-ADM-077	Upload valid CSV	1. Upload CSV with valid certificate data	Preview page shows parsed data	
TC-ADM-078	Preview import data	1. Review preview table	Shows what will be created/updated with validation status per row	
TC-ADM-079	Apply import	1. Click Apply on preview	Certificates created/updated, success notification with count	
TC-ADM-080	Cancel import	1. Click Cancel on preview	No changes made, return to import page	
TC-ADM-081	Upload invalid CSV format	1. Upload CSV with wrong column names	Error message about invalid format	

5.14 Admin-Specific Certificate Operations

ID	Description	Steps	Expected Result	Status
TC-ADM-082	Admin can delete certificates	1. View certificate 2. Click Delete 3. Confirm modal	Confirmation modal, then certificate deleted, success toast	
TC-ADM-083	Admin sees all 14 statuses	1. Open Create/Edit certificate 2. Check Status dropdown	All 14 statuses available	
TC-ADM-084	Delete notification pattern	1. Click Delete on any entity	Confirmation modal BEFORE delete, Toast AFTER success (NOT native alert/confirm)	

5.15 Delete with Dependencies

ID	Description	Steps	Expected Result	Status
TC-ADM-085	Delete company with certificates	1. Try to delete a company that has certificates	Error or warning: cannot delete	
TC-ADM-086	Delete product with certificates	1. Try to delete a product that has certificates	Error or warning: cannot delete	
TC-ADM-087	Delete cert type with certificates	1. Try to delete a certification type that has certificates	Error or warning: cannot delete	
TC-ADM-088	Delete tag with assignments	1. Try to delete a tag assigned to certificates	Tag deleted or warning about existing assignments	
TC-ADM-089	Delete role with users assigned	1. Try to delete a role that has users	Error: cannot delete role with assigned users	

5.16 Duplicate Prevention

ID	Description	Steps	Expected Result	Status
TC-ADM-090	Duplicate tag name prevention	1. Create tag "TestTag" 2. Try to create another "TestTag"	Error: Tag name must be unique	
TC-ADM-091	Duplicate company name prevention	1. Create company "TestCo" 2. Try to create another "TestCo"	Error or warning about duplicate name	
TC-ADM-092	Duplicate cert type name prevention	1. Create cert type with existing name	Error or handled gracefully	

9. Module 6: Certificate 14-State Workflow

6.1 Workflow State Definitions

#	State	Display Name	Description	Final?
1	Draft	Draft	Initial state, not yet submitted	No
2	PendingReview	Pending Review	Awaiting initial review	No
3	UnderReview	Under Review	Currently being reviewed	No
4	Approved	Approved	Approved, ready for activation	No
5	Active	Active	Currently active and valid	No

#	State	Display Name	Description	Final?
6	Suspended	Suspended	Temporarily suspended	No
7	Revoked	Revoked	Certificate revoked (near-final)	Near-final
8	Expired	Expired	Past expiration date	No
9	PendingRenewal	Pending Renewal	Within 30 days of expiration	No
10	Renewed	Renewed	Renewed, will transition to Active	No
11	Cancelled	Cancelled	Cancelled permanently	**Yes (Final)**
12	Archived	Archived	Archived permanently	**Yes (Final)**
13	OnHold	On Hold	Temporarily on hold	No
14	Rejected	Rejected	Review rejected	No

6.2 Valid State Transitions Test Matrix

ID	From State	To State	Valid?	Role	Expected Result	Status
TC-WF-001	Draft	Pending Review	Yes	Analyst/Admin	Status changes to Pending Review	
TC-WF-002	Draft	Cancelled	Yes	Admin	Status changes to Cancelled	
TC-WF-003	Draft	Active	**NO**	Any	Active NOT available from Draft	
TC-WF-004	Pending Review	Under Review	Yes	Analyst/Admin	Status updated	
TC-WF-005	Pending Review	Draft	Yes	Analyst/Admin	Status reverted	
TC-WF-006	Pending Review	Cancelled	Yes	Admin	Cancelled	
TC-WF-007	Pending Review	On Hold	Yes	Admin	Status = On Hold	
TC-WF-008	Under Review	Approved	Yes	Analyst/Admin	Status = Approved	
TC-WF-009	Under Review	Pending Review	Yes	Analyst/Admin	Status = Pending Review	
TC-WF-010	Under Review	Rejected	Yes	Admin	Status = Rejected	
TC-WF-011	Under Review	On Hold	Yes	Admin	Status = On Hold	
TC-WF-012	Approved	Active	Yes	Analyst/Admin	Status = Active	
TC-WF-013	Approved	Under Review	Yes	Analyst/Admin	Status = Under Review	
TC-WF-014	Approved	On Hold	Yes	Admin	Status = On Hold	
TC-WF-015	Active	Suspended	Yes	Admin	Status = Suspended	
TC-WF-016	Active	Expired	Yes	Admin/System	Status = Expired	
TC-WF-017	Active	Pending Renewal	Yes	Admin/System	Status = Pending Renewal	
TC-WF-018	Active	Revoked	Yes	Admin	Status = Revoked	
TC-WF-019	Active	Archived	Yes	Admin	Status = Archived	
TC-WF-020	Suspended	Active	Yes	Admin	Status = Active	
TC-WF-021	Suspended	Revoked	Yes	Admin	Status = Revoked	
TC-WF-022	Suspended	Cancelled	Yes	Admin	Status = Cancelled	

ID	From State	To State	Valid?	Role	Expected Result	Status
TC-WF-023	Suspended	Archived	Yes	Admin	Status = Archived	
TC-WF-024	Revoked	Archived	Yes	Admin	Status = Archived	
TC-WF-025	Revoked	Active	**NO**	Any	Active NOT available (near-final)	
TC-WF-026	Expired	Pending Renewal	Yes	Analyst/Admin	Status = Pending Renewal	
TC-WF-027	Expired	Archived	Yes	Admin	Status = Archived	
TC-WF-028	Expired	Active	Yes	Admin	Status = Active	
TC-WF-029	Pending Renewal	Under Review	Yes	Analyst/Admin	Status = Under Review	
TC-WF-030	Pending Renewal	Renewed	Yes	Analyst/Admin	Status = Renewed	
TC-WF-031	Pending Renewal	Expired	Yes	Admin/System	Status = Expired	
TC-WF-032	Pending Renewal	Cancelled	Yes	Admin	Status = Cancelled	
TC-WF-033	Renewed	Active	Yes	Analyst/Admin	Status = Active	
TC-WF-034	Cancelled	ANY	**NO**	Any	NO transitions available (Final state)	
TC-WF-035	Archived	ANY	**NO**	Any	NO transitions available (Final state)	
TC-WF-036	On Hold	Pending Review	Yes	Admin	Status = Pending Review	
TC-WF-037	On Hold	Under Review	Yes	Admin	Status = Under Review	
TC-WF-038	On Hold	Approved	Yes	Admin	Status = Approved	
TC-WF-039	On Hold	Cancelled	Yes	Admin	Status = Cancelled	
TC-WF-040	Rejected	Draft	Yes	Analyst/Admin	Status = Draft	
TC-WF-041	Rejected	Archived	Yes	Admin	Status = Archived	

6.3 Role-Based Status Access

ID	Description	Steps	Expected Result	Status
TC-WF-042	Admin sees all 14 statuses	1. Login as admin 2. Edit certificate 3. Check status dropdown	All 14 statuses visible	
TC-WF-043	Analyst limited statuses	1. Login as analyst 2. Edit certificate 3. Check dropdown	12 of 14 statuses (all except Rejected and Revoked)	
TC-WF-044	External cannot change status	1. Login as external 2. Verify no edit access	Cannot edit certificates at all	

6.4 Automated Workflow

ID	Description	Steps	Expected Result	Status
TC-WF-045	Auto Pending Renewal (30 days)	1. Create cert with expiration 30 days out 2. Set Active 3. Trigger background job	Status auto-changes to Pending Renewal	
TC-WF-046	Auto Expired	1. Create cert with past expiration 2. Set Active 3. Trigger check	Status auto-changes to Expired	
TC-WF-047	Status history audit trail	1. Change cert through multiple statuses 2. Check audit log	Each transition logged with: user, timestamp, from/to status	

10. Module 7: Partner API Flows

Tested using curl, PowerShell, or API testing tools (Postman, etc.)

7.1 API Authentication

ID	Description	Steps	Expected Result	Status
TC-API-001	Valid API key auth	1. Send GET /api/v1/certificates with valid X-API-Key header	200 OK with certificate data	
TC-API-002	Missing API key	1. Send GET without X-API-Key	401 Unauthorized	
TC-API-003	Invalid API key	1. Send GET with wrong key	401 Unauthorized	
TC-API-004	Revoked API key	1. Use a previously revoked API key	401 Unauthorized	

7.2 Certificate Endpoints

ID	Description	Steps	Expected Result	Status
TC-API-005	List certificates (paginated)	GET /api/v1/certificates?pageSize=10&pageNumber=1	JSON with pagination metadata + data array	
TC-API-006	Get by ID	GET /api/v1/certificates/{id}	Single certificate JSON	
TC-API-007	Get by certificate number	GET /api/v1/certificates/number/{number}	Single certificate by number	
TC-API-008	Include product IDs	GET /api/v1/certificates?includeProductIds=true	Response includes productIds array	
TC-API-009	Filter by status	GET /api/v1/certificates?status=Active	Only Active certificates returned	
TC-API-010	Filter by company	GET /api/v1/certificates?companyId={id}	Only specified company's certs	
TC-API-011	Filter by partner	GET /api/v1/certificates?partnerId={id}	Only specified partner's certs	
TC-API-012	Search by product ID	GET /api/v1/certificates/search/productid?productId={ASIN}&type=ASIN	Certificates matching product ID	
TC-API-013	Download PDF	GET /api/v1/certificates/{number}/download	PDF file returned	
TC-API-014	Non-existent certificate	GET /api/v1/certificates/999999	404 Not Found	

7.3 API Response Format

ID	Description	Steps	Expected Result	Status
TC-API-015	Pagination metadata	1. Request list endpoint	Response contains: totalCount, pageNumber, pageSize, totalPages, hasPrevious, hasNext	
TC-API-016	Certificate fields	1. Get single certificate	Contains: certificateNumber, status, issueDate, expirationDate, company, productName, certificationType, certificationTypeSubcategory, issuer, certificationResults, partner, amazonCode, pathway	

ID	Description	Steps	Expected Result	Status
TC-API-016a	Amazon Code in API response	1. Get certificate whose cert type has AmazonCode set	Response includes amazonCode field with correct mapping value	
TC-API-016b	Pathway in API response	1. Get certificate with Pathway set	Response includes pathway field with selected pathway name	
TC-API-016c	Amazon Code in GS1 JSON-LD response	1. Get certificate via GS1 JSON-LD endpoint	amazonCode field present in GS1 formatted response	
TC-API-016d	Pathway in GS1 JSON-LD response	1. Get certificate via GS1 JSON-LD endpoint	pathway field present in GS1 formatted response	
TC-API-016e	Amazon Code null when not set	1. Get certificate whose cert type has no AmazonCode	amazonCode field is null or omitted (no error)	
TC-API-016f	Pathway null when not set	1. Get certificate with no Pathway selected	pathway field is null or omitted (no error)	
TC-API-017	Issuer always "Green Circle Certified"	1. Check issuer field on any cert	issuer = "Green Circle Certified" (static value)	
TC-API-018	Product ID format	1. Get cert with includeProductIds=true	Each productId has: type, value, status, partner, productName, productSku, description	
TC-API-019	Date format ISO 8601	1. Check date fields	Dates in format: YYYY-MM-DDTHH:mm:ssZ	

7.4 Rate Limiting

ID	Description	Steps	Expected Result	Status
TC-API-020	Rate limit enforced	1. Send rapid requests exceeding rate limit	429 Too Many Requests after limit exceeded	
TC-API-021	Rate limit headers	1. Check response headers	Headers include rate limit info (remaining, reset)	

11. Module 8: Cross-Role Permission Matrix Validation

8.1 Page Access by Role

ID	URL	Public	External	Analyst	Admin	Status
TC-PERM-001	/Certificate/CertificateTileView	Allowed	Allowed	Allowed	Allowed	
TC-PERM-002	/Certificate/CertificateTileViewExternal	Denied	Allowed	Allowed	Allowed	
TC-PERM-003	/Certificate/CertificateTileViewInternal	Denied	Denied	Allowed	Allowed	
TC-PERM-004	/Certificate/Create	Denied	Denied	Allowed	Allowed	
TC-PERM-005	/Certificate/Edit/{number}	Denied	Denied	Allowed	Allowed	

ID	URL	Public	External	Analyst	Admin	Status
TC-PERM-006	/Dashboard/Index	Denied	Denied	Allowed	Allowed	
TC-PERM-007	/Company/Index	Denied	Denied	Allowed	Allowed	
TC-PERM-008	/Product/Management	Denied	Allowed (Read-only)	Allowed	Allowed	
TC-PERM-009	/Admin/UserManagement	Denied	Denied	Denied	Allowed	
TC-PERM-010	/Admin/RoleManagement	Denied	Denied	Denied	Allowed	
TC-PERM-011	/Admin/CertificateTypes	Denied	Denied	Denied	Allowed	
TC-PERM-012	/Admin/CertificateFields	Denied	Denied	Denied	Allowed	
TC-PERM-013	/Admin/Tags	Denied	Denied	Denied	Allowed	
TC-PERM-014	/Admin/Partners	Denied	Denied	Denied	Allowed	
TC-PERM-015	/Admin/Pathways	Denied	Denied	Denied	Allowed	
TC-PERM-016	/Admin/ProductTypes	Denied	Denied	Denied	Allowed	
TC-PERM-017	/Admin/PartnerAPI	Denied	Denied	Denied	Allowed	
TC-PERM-018	/Admin/GoogleAnalytics	Denied	Denied	Denied	Allowed	
TC-PERM-019	/CertificateImport/Index	Denied	Denied	**Denied**	Allowed	

8.2 Feature Access by Role

ID	Feature	Public	External	Analyst	Admin	Status
TC-PERM-020	View certificates	Yes (public only)	Yes (own company)	Yes (all)	Yes (all)	
TC-PERM-021	Create certificates	No	No	Yes	Yes	
TC-PERM-022	Edit certificates	No	No	Yes	Yes	
TC-PERM-023	Delete certificates	No	No	No	Yes	
TC-PERM-024	Manage workflow	No	No	Yes (limited)	Yes (full)	
TC-PERM-025	Export data (CSV)	No	Yes (own data)	Yes	Yes	
TC-PERM-026	Import data - Certificate Import	No	No	**No (Admin only)**	Yes	
TC-PERM-026a	Import data - Product IDs within cert form	No	No	Yes	Yes	
TC-PERM-027	View internal tags	No	No	Yes	Yes	
TC-PERM-028	View external tags	No	Yes	Yes	Yes	
TC-PERM-029	Manage users	No	No	No	Yes	
TC-PERM-030	Manage roles	No	No	No	Yes	
TC-PERM-031	Manage companies	No	No	No	Yes	
TC-PERM-032	Manage tags	No	No	No	Yes	
TC-PERM-033	Manage partners	No	No	No	Yes	
TC-PERM-034	API access	No	No	No	Yes	
TC-PERM-035	View products	No	Yes (own company)	Yes	Yes	

ID	Feature	Public	External	Analyst	Admin	Status
TC-PERM-036	Manage products	No	No	No	Yes	
TC-PERM-037	System settings	No	No	No	Yes	

8.3 UI Element Visibility by Role

ID	UI Element	Public	External	Analyst	Admin	Status
TC-PERM-038	Navigation: Administration menu	Hidden	Hidden	Hidden	Visible	
TC-PERM-039	Navigation: Dashboard	Hidden	Hidden	Visible	Visible	
TC-PERM-040	Navigation: Companies & Products	Hidden	Products only	Both	Both	
TC-PERM-041	Add Certificate button	Hidden	Hidden	Visible	Visible	
TC-PERM-042	Edit Certificate button	Hidden	Hidden	Visible	Visible	
TC-PERM-043	Delete Certificate button	Hidden	Hidden	Hidden	Visible	
TC-PERM-044	Internal Tags column	Hidden	Hidden	Visible	Visible	
TC-PERM-045	Product Actions column	Hidden	Hidden	Visible	Visible	
TC-PERM-046	Import/Export buttons (Admin pages)	Hidden	Hidden	Hidden	Visible	

12. Module 9: Data Export & Import

9.1 CSV Export

ID	Description	Steps	Expected Result	Status
TC-EXP-001	Export certificates	1. Navigate to cert view (Analyst+) 2. Click Export	CSV file downloads with certificate data	
TC-EXP-002	Export users	1. Admin > User Management 2. Click Export	CSV downloads with user data	
TC-EXP-003	Export roles	1. Admin > Role Management 2. Click Export	CSV downloads with role data	
TC-EXP-004	Export tags	1. Admin > Tags 2. Click Export	CSV downloads with tag data	
TC-EXP-005	Export partners	1. Admin > Partners 2. Click Export	CSV downloads with partner data	
TC-EXP-006	Export products	1. Product Management 2. Click Export	CSV downloads with product data	
TC-EXP-007	Export companies	1. Company Management 2. Click Export	CSV downloads with company data	
TC-EXP-008	CSV format only	1. Check any export	File is .csv format (NOT .xlsx, .json, or .pdf)	
TC-EXP-009	Export with filters applied	1. Apply filters 2. Click Export	Only filtered data exported	
TC-EXP-010	External export own data only	1. Login as external 2. Export	Only own company's data in export	

9.2 CSV Import

ID	Description	Steps	Expected Result	Status
TC-IMP-001	Download template - Users	1. Admin > User Management 2. Click Template	CSV template with correct column headers	
TC-IMP-002	Download template - Tags	1. Admin > Tags 2. Click Template	CSV template downloads	
TC-IMP-003	Download template - Companies	1. Company page 2. Click Template	CSV template downloads	
TC-IMP-004	Import valid CSV	1. Click Import 2. Upload properly formatted CSV	Data imported successfully	
TC-IMP-005	Import with validation errors	1. Upload CSV with invalid data	Error report showing which rows failed and why	
TC-IMP-006	Import duplicate records	1. Upload CSV with existing data	Handled gracefully (update or skip with message)	
TC-IMP-007	Partners page - No Import	1. Go to Partners page	Import button does NOT exist	
TC-IMP-008	Partners page - No Template	1. Go to Partners page	Template button does NOT exist	

13. Module 10: Edge Cases & Negative Testing

10.1 Form Validation

ID	Description	Steps	Expected Result	Status
TC-NEG-001	Certificate - empty required fields	1. Create cert with no fields 2. Submit	Validation errors for all required fields	
TC-NEG-002	Certificate - duplicate number	1. Create cert with existing number	Error: Certificate number must be unique	
TC-NEG-003	User - empty email	1. Create user with no email 2. Submit	Validation error for email	
TC-NEG-004	User - invalid email format	1. Create user with "notanemail" 2. Submit	Validation error for email format	
TC-NEG-005	User - duplicate email	1. Create user with existing email	Error: Email must be unique	
TC-NEG-006	Role - empty name	1. Create role with no name	Validation error	
TC-NEG-007	Tag - empty name	1. Create tag with no name	Validation error	
TC-NEG-008	Company - empty name	1. Create company with no name	Validation error	
TC-NEG-009	Product ID - invalid type	1. Add product ID without selecting type	Validation error	
TC-NEG-010	Product ID - empty value	1. Add product ID without value	Validation error	

10.2 Notification Pattern Validation

> **Critical**: NO native browser alert(), confirm(), or prompt() should appear anywhere in the system.

ID	Description	Steps	Expected Result	Status
TC-NEG-011	Delete cert notification	1. Delete a certificate	Confirmation modal BEFORE, toast AFTER	
TC-NEG-012	Delete user notification	1. Delete a user	Confirmation modal BEFORE, toast AFTER	
TC-NEG-013	Delete tag notification	1. Delete a tag	Confirmation modal BEFORE, toast AFTER	
TC-NEG-014	Delete role notification	1. Delete a role	Confirmation modal BEFORE, toast AFTER	
TC-NEG-015	Create success notification	1. Create any entity	Toast notification for success (no modal)	
TC-NEG-016	Edit success notification	1. Edit any entity	Toast notification for success	
TC-NEG-017	No native alerts anywhere	1. Use every feature in the system	NEVER see a native browser alert/confirm/prompt	

10.3 Data Boundary Testing

ID	Description	Steps	Expected Result	Status
TC-NEG-018	Certification Results max length	1. Enter 1001+ characters in Certification Results	Truncated or rejected at 1000 chars	
TC-NEG-019	Certificate number format	1. Verify auto-generated number format	Format: YY-NNNN (e.g., 24-0366)	
TC-NEG-020	Very long company name	1. Create company with 255+ character name	Handled gracefully	
TC-NEG-021	Special characters in fields	1. Enter <script>alert('xss')</script> in text fields	HTML encoded, no script execution (XSS prevention)	
TC-NEG-022	SQL injection attempt	1. Enter ' OR 1=1 -- in search fields	No SQL injection, treated as literal text	
TC-NEG-023	Large CSV import	1. Import CSV with 1000+ rows	Handled gracefully	

10.4 Browser & Responsiveness

ID	Description	Steps	Expected Result	Status
TC-NEG-024	Minimum font size	1. Inspect any page text	All text >= 16px	
TC-NEG-025	Mobile responsive tile view	1. Resize browser to mobile width	Tile view adapts, remains usable	
TC-NEG-026	Admin page responsive	1. Resize admin page to tablet width	Table scrolls or adapts	

14. Module 10b: Audit Trail & Change History

Purpose: Verify the audit trail system captures all changes to entities, tracks certificate status transitions, and records user/session context.

10b.1 Field-Level Change Tracking

ID	Description	Steps	Expected Result	Status
TC-AUDIT-001	Field changes logged in AuditHistory	1. Edit a certificate 2. Query AuditHistory table	Records show EntityType="Certificate", FieldName, OldValue, NewValue	
TC-AUDIT-003	ChangedBy captures username	1. Login as analyst 2. Edit a certificate 3. Query AuditHistory	ChangedBy field contains "analyst" username	
TC-AUDIT-004	IP address and User-Agent logged	1. Make any change 2. Query AuditHistory	IpAddress and UserAgent fields are populated	
TC-AUDIT-006	Change source tracked	1. Edit via UI 2. Import via CSV 3. Query AuditHistory	UI: ChangeSource="UI", Import: ChangeSource="Import"	

10b.2 Certificate Status Transition History

ID	Description	Steps	Expected Result	Status
TC-AUDIT-002	Status transitions logged	1. Change cert from Draft to Pending Review 2. Query CertificateStatusHistory	Record with PreviousStatus="Draft", NewStatus="PendingReview", timestamp, ChangedBy	
TC-AUDIT-005	Import batch correlation	1. Import certificates via CSV 2. Query AuditHistory	All records from same import share same ImportBatchId	
TC-AUDIT-009	Automated vs manual transition flag	1. Manually change status 2. Check auto-expiration 3. Query history	Manual: IsAutomatedTransition=false. Automated: IsAutomatedTransition=true	

10b.3 Audit Trail UI (GAP - Not Yet Built)

ID	Description	Steps	Expected Result	Status
TC-AUDIT-007	Status history on certificate detail page	1. Open certificate detail page for cert with multiple status changes	Status history timeline/table visible	Gap - UI not built
TC-AUDIT-008	Admin audit log viewer	1. Navigate to Admin > Audit Log 2. Search by entity type, date, user	Searchable/filterable audit log with CSV export	Gap - UI not built

Module Total: 9 test cases (7 backend-verifiable, 2 UI gaps)

PART B: END-TO-END BUSINESS SCENARIOS

These scenarios simulate real-world business journeys that cross multiple roles, pages, and data dependencies. Each scenario must be executed **in sequence** - steps build on each other.

Module 11: E2E Scenario 1 - Certificate Full Lifecycle

Business Story: A new certification is created by an Analyst, reviewed and approved by an Admin, made visible to the public and partners, viewed by an External client, and eventually expires and gets archived.

Roles: Admin, Analyst, External, Public, Partner (API) | **Time**: 30-45 min

Scenario Steps

Step	ID	Role	Action	Expected Result	Status
1	E2E-LC-001	Analyst	Login as analyst	Redirected to internal tile view or dashboard	
2	E2E-LC-002	Analyst	Click "Create Certificate"	Create form loads, cert number auto-generated	
3	E2E-LC-003	Analyst	Select Company from dropdown	Company selected, products may filter	
4	E2E-LC-004	Analyst	Select Product from dropdown	Product selected	
5	E2E-LC-005	Analyst	Select Certification Type (parent)	Subcategory dropdown populates	
6	E2E-LC-006	Analyst	Select Subcategory	Custom fields load dynamically	
7	E2E-LC-007	Analyst	Fill custom fields (if any)	Fields accept valid input	
8	E2E-LC-008	Analyst	Set Effective Date = today	Date accepted	
9	E2E-LC-009	Analyst	Set Expiration Date = 1 year from today	Date accepted (after effective date)	
10	E2E-LC-010	Analyst	Set Location and Country	Fields populated	
11	E2E-LC-011	Analyst	Select Partner = "Amazon"	Partner set	
12	E2E-LC-012	Analyst	Enter Certification Results = "95.5% recycled content"	Text accepted in optional field	
13	E2E-LC-013	Analyst	Upload PDF certificate (valid < 10MB)	File preview generated, success toast	
14	E2E-LC-014	Analyst	Add External Tag ("Sustainable")	Tag badge appears	
15	E2E-LC-015	Analyst	Add Internal Tag ("PriorityReview")	Tag badge appears	
16	E2E-LC-016	Analyst	Add Product ID: Type=ASIN, Value=B08X7N9MRQ	Product ID appears in table	
17	E2E-LC-017	Analyst	Add Product ID: Type=UPC, Value=846378985421	Second product ID in table	
18	E2E-LC-018	Analyst	Set Status = "Draft"	Status set to Draft	
19	E2E-LC-019	Analyst	Click Save	Certificate created, redirected to Details, success toast. **Record Certificate Number.**	
20	E2E-LC-020	Analyst	Verify Details page shows all entered data	All fields match entered data	
21	E2E-LC-021	Analyst	Verify Certificate Number displayed (NOT database ID)	Shows YY-NNNN format in heading and URL	

Step	ID	Role	Action	Expected Result	Status
22	E2E-LC-022	Public	Open incognito, go to public tile view	Page loads	
23	E2E-LC-023	Public	Search for the new certificate number	Certificate NOT found (Draft, not visible to public)	
24	E2E-LC-024	Analyst	Edit certificate, change Status to "Pending Review"	Status updated, save successful	
25	E2E-LC-025	Admin	Login as admin	Dashboard loads	
26	E2E-LC-026	Admin	Find certificate in internal tile view	Certificate visible with "Pending Review" status badge	
27	E2E-LC-027	Admin	Edit: change Status to "Under Review"	Status = Under Review	
28	E2E-LC-028	Admin	Edit: change Status to "Approved"	Status = Approved	
29	E2E-LC-029	Admin	Edit: change Status to "Active"	Status = Active. Certificate is now LIVE.	
30	E2E-LC-030	Public	Refresh public tile view, search for certificate	Certificate NOW appears (Active)	
31	E2E-LC-031	Public	Click on the certificate tile	Detail view shows data. External tags visible. Internal tags NOT visible. No Edit/Delete.	
32	E2E-LC-032	Public	Click Download PDF	PDF file downloads successfully	
33	E2E-LC-033	Public	Verify Certificate Number in URL (not ID)	URL shows /Certificate/CertificateSingleView/XX-XXXX	
34	E2E-LC-034	External	Login as external user (same company)	External tile view loads	
35	E2E-LC-035	External	Find certificate in external tile view	Certificate visible. External tags shown. Internal tags NOT shown. No Edit/Delete.	
36	E2E-LC-036	External	Click on certificate for details	Read-only view. Certification Results visible. No Edit/Delete.	
37	E2E-LC-037	API	GET /api/v1/certificates?includeProductIds=true	Certificate appears with certificationResults, productIds	
38	E2E-LC-038	API	Search by ASIN	Certificate returned by ASIN search	
39	E2E-LC-039	API	Download PDF via API	PDF file returned	
40	E2E-LC-040	Admin	Edit: change Status to "Suspended"	Status = Suspended	
41	E2E-LC-041	Public	Refresh public tile view, search for certificate	Certificate NO LONGER visible	
42	E2E-LC-042	Admin	Edit: change Status back to "Active"	Status = Active, certificate reappears	

Step	ID	Role	Action	Expected Result	Status
43	E2E-LC-043	Admin	Edit: change Status to "Expired"	Status = Expired	
44	E2E-LC-044	Admin	Edit: change Status to "Archived"	Status = Archived (FINAL state)	
45	E2E-LC-045	Admin	Try to edit Archived certificate status	No status transitions available. Certificate is permanently archived.	
46	E2E-LC-046	Public	Search for archived certificate	Certificate NOT visible	

Verification Checklist

- [ ] Certificate created with all fields populated
- [ ] Workflow progressed: Draft → Pending Review → Under Review → Approved → Active
- [ ] Public visibility activated only when Active
- [ ] External user saw only external tags, not internal
- [ ] Partner API returned certificate with product IDs and certification results
- [ ] Suspension hid certificate from public, reactivation restored it
- [ ] Archived certificate reached final state with no further transitions
- [ ] Certificate number (not ID) used throughout all views and URLs

Module 12: E2E Scenario 2 - New Company Onboarding

Business Story: A new client company is set up. Products are created. An external user is associated. The external user can only see their company's data.

Roles: Admin, External (new user) | **Time**: 20-30 min

Scenario Steps

Step	ID	Role	Action	Expected Result	Status
1	E2E-CO-001	Admin	Login as admin	Dashboard loads	
2	E2E-CO-002	Admin	Navigate to Company Management	Company list loads	
3	E2E-CO-003	Admin	Click Add New Company	Add form/modal opens	
4	E2E-CO-004	Admin	Enter: Name="TestCo UAT", Location, Country, Email, Website	Fields accepted	
5	E2E-CO-005	Admin	Save company	Company created, success toast	
6-10	E2E-CO-006 to 010	Admin	Create 2 products under company	Products created and linked	
11-14	E2E-CO-011 to 014	Admin	Create external user linked to company	User created with External role	
15	E2E-CO-015	Admin	Create Active certificate for TestCo UAT	Certificate created	
16	E2E-CO-016	Admin	Create Active certificate for DIFFERENT company	Second certificate created	
17	E2E-CO-017	Admin	Logout	Session ended	

Step	ID	Role	Action	Expected Result	Status
18	E2E-CO-018	External	Login as new external user	Redirected to External Tile View	
19	E2E-CO-019	External	View certificate tile list	ONLY TestCo UAT certificate visible	
20	E2E-CO-020	External	Search for other company's cert	No results. Cannot see other company data.	
21	E2E-CO-021	External	Navigate to Products	ONLY own company's products visible	
22	E2E-CO-022	External	Verify no Actions column	No Edit/Delete buttons. Read-only.	
23	E2E-CO-023	External	Verify no Admin menu	No Administration menu	
24	E2E-CO-024	External	Navigate to /Admin/UserManagement	Access denied	
25	E2E-CO-025	External	Navigate to /Certificate/Create	Access denied	
26	E2E-CO-026	External	Export certificates	CSV contains ONLY own company certificates	

Module 13: E2E Scenario 3 - Partner API Integration

Business Story: An admin configures a partner, generates an API key, and the partner uses the API to discover certificates, search by product ID, and download PDFs.

Roles: Admin, Partner (API) | **Time**: 20-30 min | **Tool**: curl/PowerShell/Postman

Scenario Steps

Step	ID	Role	Action	Expected Result	Status
1	E2E-API-001	Admin	Navigate to Partner API Management	API management page loads	
2	E2E-API-002	Admin	Generate new API key for "Amazon"	Key generated. **Copy and save.**	
3	E2E-API-003	Partner	Test auth: GET /api/v1/certificates with X-API-Key	200 OK with JSON response	
4	E2E-API-004	Partner	Test NO auth: GET without key	401 Unauthorized	
5	E2E-API-005	Partner	List page 1: ?pageNumber=1&pageSize=5	Pagination metadata + data array	
6	E2E-API-006	Partner	Verify pagination: check hasNext, request page 2	Page 2 returns next set	
7	E2E-API-007	Partner	Get single cert by number	Single certificate object with all fields	
8	E2E-API-008	Partner	Verify response fields	All fields present. issuer = "Green Circle Certified"	
9	E2E-API-009	Partner	Request with includeProductIds=true	productIds array populated	
10	E2E-API-010	Partner	Search by ASIN	Certificate(s) with matching ASIN returned	
11	E2E-API-011	Partner	Search by UPC	Certificate(s) with matching UPC returned	
12	E2E-API-012	Partner	Filter by status=Active	Only Active certificates returned	
13	E2E-API-013	Partner	Download PDF	PDF file with correct content-type	
14	E2E-API-014	Partner	Request non-existent certificate	404 Not Found	
15	E2E-API-015	Partner	Verify date format	ISO 8601: YYYY-MM-DDTHH:mm:ssZ	
16	E2E-API-016	Partner	Send rapid requests (exceed rate limit)	429 Too Many Requests	

Step	ID	Role	Action	Expected Result	Status
17	E2E-API-017	Admin	Revoke the API key	Key marked as revoked	
18	E2E-API-018	Partner	Attempt request with revoked key	401 Unauthorized	

Module 14: E2E Scenario 4 - Admin System Setup (Day 1)

Business Story: A brand-new system is configured. Admin sets up all master data, then an Analyst creates a certificate using all the configured data.

Roles: Admin, Analyst | **Time**: 30-40 min

Scenario Steps

Step	ID	Role	Action	Expected Result	Status
Phase 1: Master Data Setup (Admin)					
1	E2E-SETUP-001	Admin	Login as admin	Dashboard loads	
2	E2E-SETUP-002	Admin	Create Cert Type: Parent="Environmental Compliance"	Parent type created	
3	E2E-SETUP-003	Admin	Create Subcategory: "Water Conservation Level 1"	Child type created under parent	
4	E2E-SETUP-004	Admin	Create Subcategory: "Water Conservation Level 2"	Second child type created	
5	E2E-SETUP-005	Admin	Set AmazonCode on parent cert type	AmazonCode saved	
6	E2E-SETUP-006	Admin	Create Custom Field: "Testing Lab", Text, Required	Custom field created and linked	
7	E2E-SETUP-007	Admin	Create Custom Field: "Compliance Score", Number, Optional	Second custom field created	
8	E2E-SETUP-008	Admin	Create External Tag: "Water Positive"	External tag created	
9	E2E-SETUP-009	Admin	Create Internal Tag: "FastTrack"	Internal tag created	
10	E2E-SETUP-010	Admin	Create/verify Partner	Partner created/verified	
11	E2E-SETUP-011	Admin	Create Product Type hierarchy	Three-level hierarchy created	
12	E2E-SETUP-012	Admin	Create Company: "AquaTech Solutions"	Company created	
13	E2E-SETUP-013	Admin	Create Product: "AquaFilter Pro"	Product created and linked	
14	E2E-SETUP-014	Admin	Logout	Session ended	
Phase 2: Certificate Creation Using Setup Data (Analyst)					

Step	ID	Role	Action	Expected Result	Status
15	E2E-SETUP-015	Analyst	Login as analyst	Internal view loads	
16	E2E-SETUP-016	Analyst	Navigate to Create Certificate	Form loads	
17	E2E-SETUP-017	Analyst	Select Company = "AquaTech Solutions"	Company available in dropdown	
18	E2E-SETUP-018	Analyst	Select Product = "AquaFilter Pro"	Product available (linked to company)	
19	E2E-SETUP-019	Analyst	Select Cert Type = "Environmental Compliance"	Subcategory dropdown populates	
20	E2E-SETUP-020	Analyst	Select Subcategory = "Water Conservation Level 1"	Custom fields appear: "Testing Lab" + "Compliance Score"	
21	E2E-SETUP-021	Analyst	Fill "Testing Lab" = "National Water Testing Institute"	Required field accepts input	
22	E2E-SETUP-022	Analyst	Fill "Compliance Score" = "94.7"	Optional number field accepts input	
23	E2E-SETUP-023	Analyst	Try Save WITHOUT "Testing Lab" (clear it)	Validation error: "Testing Lab is required"	
24	E2E-SETUP-024	Analyst	Refill, Add External Tag "Water Positive"	Tag from admin setup appears and selected	
25	E2E-SETUP-025	Analyst	Add Internal Tag "FastTrack"	Internal tag available and selected	
26	E2E-SETUP-026	Analyst	Set dates, location, partner. Click Save.	Certificate created with ALL admin-configured data	
27	E2E-SETUP-027	Analyst	View Details page	Verify all data correct	

Module 15: E2E Scenario 5 - Certificate Renewal Lifecycle

Business Story: An active certificate approaches expiration. The system auto-transitions to Pending Renewal. Analyst submits for review, it gets renewed and returns to Active.

Roles: Admin, Analyst, System | **Time**: 20 min

Step	ID	Role	Action	Expected Result	Status
1	E2E-REN-001	Admin	Create cert Active, Expiration = today + 25 days	Certificate created as Active	
2	E2E-REN-002	System	Trigger expiration check	Auto-transitions to Pending Renewal (30-day window)	
3	E2E-REN-003	Analyst	View in internal tile view	"Pending Renewal" status badge	
4	E2E-REN-004	Analyst	Verify "Expiring Soon" indicator	Visual indicator showing cert is expiring soon	

Step	ID	Role	Action	Expected Result	Status
5	E2E-REN-005	Analyst	Edit: change to "Under Review"	Status = Under Review	
6	E2E-REN-006	Admin	Change status to "Renewed"	Status = Renewed	
7	E2E-REN-007	Admin	Change from Renewed to "Active"	Status = Active (renewed and active)	
8	E2E-REN-008	Admin	Verify expiration date updated	Expiration date updated for renewed cert	
9	E2E-REN-009	Public	Verify certificate visible	Certificate appears as Active	
10	E2E-REN-010	Admin	Check status history	Full history: Active → Pending Renewal → Under Review → Renewed → Active	

Alternative Path: Expiration Without Renewal

Step	ID	Role	Action	Expected Result	Status
11	E2E-REN-011	Admin	Create cert Active, Expiration = yesterday	Certificate created	
12	E2E-REN-012	System	Trigger expiration check	Auto-transitions to Expired	
13	E2E-REN-013	Public	Search for expired certificate	Certificate NOT visible	
14	E2E-REN-014	Admin	Archive expired certificate	Status = Archived (final state)	

Module 16: E2E Scenario 6 - Rejection & Recovery Flow

Business Story: An Analyst creates a certificate that gets rejected. The Analyst corrects issues and resubmits. Certificate eventually gets approved and activated.

Roles: Analyst, Admin | **Time**: 15-20 min

Step	ID	Role	Action	Expected Result	Status
1	E2E-REJ-001	Analyst	Create cert with incomplete data (missing PDF)	Certificate created in Draft	
2	E2E-REJ-002	Analyst	Change to Pending Review	Status = Pending Review	
3	E2E-REJ-003	Admin	Review certificate, identify issues	Admin sees incomplete cert	
4	E2E-REJ-004	Admin	Change to Under Review	Status = Under Review	
5	E2E-REJ-005	Admin	Reject: change to "Rejected"	Status = Rejected	
6	E2E-REJ-006	Analyst	View rejected certificate	"Rejected" status shown	
7	E2E-REJ-007	Analyst	Verify available transitions from Rejected	Can go to: Draft or Archived only	

Step	ID	Role	Action	Expected Result	Status
8	E2E-REJ-008	Analyst	Change back to "Draft"	Status = Draft (restart)	
9	E2E-REJ-009	Analyst	Fix: Upload PDF, add missing data	Certificate now complete	
10	E2E-REJ-010	Analyst	Resubmit: change to "Pending Review"	Status = Pending Review (second attempt)	
11	E2E-REJ-011	Admin	Review: Pending Review → Under Review → Approved	Status = Approved	
12	E2E-REJ-012	Admin	Activate: Approved → Active	Status = Active. Certificate is now live.	
13	E2E-REJ-013	Public	Verify certificate in public view	Certificate visible	
14	E2E-REJ-014	Admin	Check full audit trail	History: Draft → PendingReview → UnderReview → Rejected → Draft → PendingReview → UnderReview → Approved → Active	

Module 17: E2E Scenario 7 - Bulk Operations Workflow

Business Story: Admin downloads templates, prepares bulk data, imports via CSV, previews, applies, then exports to verify.

Roles: Admin | **Time**: 25-35 min

Step	ID	Role	Action	Expected Result	Status
Phase 1: Template Download					
1	E2E-BULK-001	Admin	User Management → Template	CSV template downloads	
2	E2E-BULK-002	Admin	Tags → Template	Tag import template downloads	
3	E2E-BULK-003	Admin	Companies → Template	Company import template downloads	
4	E2E-BULK-004	Admin	Certificate Import → Template	Certificate import template downloads	
5	E2E-BULK-005	Admin	Verify all templates are CSV format	All files are .csv format	
Phase 2: Import Companies					
6	E2E-BULK-006	Admin	Fill template with 3 companies	Data prepared	
7	E2E-BULK-007	Admin	Import companies CSV	Companies imported, toast notification	
8	E2E-BULK-008	Admin	Verify all 3 companies in list	All 3 visible	
Phase 3: Import Tags					
9	E2E-BULK-009	Admin	Fill tag template: 2 external + 1 internal	Tags prepared	
10	E2E-BULK-010	Admin	Import tags CSV	Tags imported	
11	E2E-BULK-011	Admin	Filter by type to verify	Correctly typed tags	
Phase 4: Bulk Certificate Import					
12	E2E-BULK-012	Admin	Fill cert template with 5 certificates	CSV prepared	
13	E2E-BULK-013	Admin	Upload certificate CSV	File accepted, redirected to preview	

Step	ID	Role	Action	Expected Result	Status
14	E2E-BULK-014	Admin	Review import preview	Preview shows: X new, Y modified, Z errors	
15	E2E-BULK-015	Admin	Verify auto-created entities	Preview shows auto-creation of missing products	
16	E2E-BULK-016	Admin	Click Apply	Import executed, success notification	
17	E2E-BULK-017	Admin	Navigate to Internal Tile View	All 5 imported certificates visible	
Phase 5: Export & Verify					
18	E2E-BULK-018	Admin	Export certificates	CSV downloads	
19	E2E-BULK-019	Admin	Verify imported data in CSV	All 5 certificates present with correct data	
20	E2E-BULK-020	Admin	Export companies	All 3 bulk companies in export	
21	E2E-BULK-021	Admin	Export tags	Bulk-imported tags in export	
Phase 6: Negative Import Testing					
22	E2E-BULK-022	Admin	Upload CSV with duplicate cert numbers	Preview shows "Modified" or "Duplicate" status	
23	E2E-BULK-023	Admin	Upload CSV with invalid data	Preview shows error rows with messages	
24	E2E-BULK-024	Admin	Cancel import of invalid data	No changes applied	

Module 18: E2E Scenario 8 - User Access Lifecycle

Business Story: Admin creates a new Analyst user. User logs in, works normally, admin changes role, eventually deactivates account.

Roles: Admin, Analyst (new) | **Time**: 20 min

Step	ID	Role	Action	Expected Result	Status
Phase 1: Account Creation					
1-5	E2E-USER-001 to 005	Admin	Create user: "UAT Analyst", Role=Analyst, Password="Initial123!"	User created, visible in list	
Phase 2: First Login					
6	E2E-USER-006	New Analyst	Login with Initial123!	Login succeeds	
7	E2E-USER-007	New Analyst	If force password reset: enter new password	Password changed	
8	E2E-USER-008	New Analyst	If 2FA required: complete setup	2FA configured	
9-10	E2E-USER-009-010	New Analyst	Verify correct landing page and navigation	Analyst-appropriate view and nav	
Phase 3: Normal Operations					
11	E2E-USER-011	New Analyst	Create a certificate	Can create certificates	
12	E2E-USER-012	New Analyst	Edit the certificate	Can edit certificates	
13	E2E-USER-013	New Analyst	Try to delete a certificate	NO delete button visible	
14-16	E2E-USER-014 to 016	New Analyst	View Dashboard, access Companies, try Admin pages	Dashboard works. Companies work. Admin denied.	
17	E2E-USER-017	New Analyst	Logout	Session ended	

Step	ID	Role	Action	Expected Result	Status
Phase 4: Account Modification					
18-19	E2E-USER-018-019	Admin	Change user role from Analyst to External	Role changed	
20	E2E-USER-020	Now External	Login	Redirected to External Tile View	
21	E2E-USER-021	Now External	Verify navigation reflects External role	Only Certificates and Products. No Dashboard/Admin.	
22	E2E-USER-022	Now External	Try to create certificate	Access denied	
23	E2E-USER-023	Now External	Logout	Session ended	
Phase 5: Deactivation					
24-25	E2E-USER-024-025	Admin	Deactivate user	User deactivated/deleted	
26	E2E-USER-026	Deactivated	Try to login	Login fails with appropriate message	

Module 19: E2E Scenario 9 - Search & Discovery Cross-Role

Business Story: Same certificates viewed from all 4 perspectives to verify each role sees correct data, filters, and UI elements.

Roles: Public, External, Analyst, Admin | **Time**: 25-30 min

Pre-conditions

- Cert "A": Active, Partner=Amazon, Company=TestCo, ExternalTag="Sustainable", InternalTag="Priority", has ASIN
- Cert "B": Active, Partner=None, Company=OtherCo, ExternalTag="Verified"
- Cert "C": Draft, Company=TestCo
- Cert "D": Archived, Company=TestCo
- External user associated with "TestCo"

Step	ID	Role	Action	Expected Result	Status
Public View					
1	E2E-SEARCH-001	Public	Navigate to public tile view	Cert A and B visible. C (Draft) and D (Archived) NOT visible.	
2	E2E-SEARCH-002	Public	Check filter panel	External tags shown. NO internal tags.	
3	E2E-SEARCH-003	Public	Filter by tag "Sustainable"	Only Cert A shown	
4	E2E-SEARCH-004	Public	Clear filter, search by ASIN	Cert A found via autocomplete	
5	E2E-SEARCH-005	Public	Check Cert A tile details	No Edit/Delete. No internal tags. Cert Number shown.	

Step	ID	Role	Action	Expected Result	Status
6	E2E-SEARCH-006	Public	Copy shareable URL with filter	URL contains search parameters	
7	E2E-SEARCH-007	Public	Open shareable URL in incognito	Same filtered results displayed	
External View (TestCo user)					
8	E2E-SEARCH-008	External	Login as external (TestCo)	External tile view loads	
9	E2E-SEARCH-009	External	View certificate list	Cert A visible. B, C, D NOT visible.	
10	E2E-SEARCH-010	External	Check filter panel	External tags only. No internal tags.	
11	E2E-SEARCH-011	External	Search for Cert B by number	No results (different company)	
12	E2E-SEARCH-012	External	View Cert A details	External tags visible. Internal tags NOT visible. No Edit/Delete.	
13	E2E-SEARCH-013	External	Check Products page	Only TestCo products. No Actions column.	
Analyst View					
14	E2E-SEARCH-014	Analyst	Login as analyst	Internal tile view loads	
15	E2E-SEARCH-015	Analyst	View certificate list	ALL 4 certs visible across all statuses/companies	
16	E2E-SEARCH-016	Analyst	Check filter panel	Both internal AND external tags. All companies/statuses.	
17	E2E-SEARCH-017	Analyst	Filter by internal tag "Priority"	Only Cert A shown	
18	E2E-SEARCH-018	Analyst	Filter by status "Draft"	Only Cert C shown	
19	E2E-SEARCH-019	Analyst	View Cert A details	ALL data visible. Edit button. No Delete button.	
20	E2E-SEARCH-020	Analyst	Click Edit on Cert A	Edit page loads with all data	
Admin View					
21	E2E-SEARCH-021	Admin	Login as admin	Dashboard or internal tile view	
22	E2E-SEARCH-022	Admin	View certificate list	ALL 4 certs visible	
23	E2E-SEARCH-023	Admin	View Cert A details	ALL data. Edit AND Delete buttons.	
24	E2E-SEARCH-024	Admin	Filter by status "Archived"	Only Cert D shown	

Step	ID	Role	Action	Expected Result	Status
25	E2E-SEARCH-025	Admin	Check status dropdown for Cert C (Draft)	Valid transitions available	
26	E2E-SEARCH-026	Admin	Check status dropdown for Cert D (Archived)	No transitions (final state)	

Cross-Role Comparison Summary

Aspect	Public	External (TestCo)	Analyst	Admin
Cert A (Active, TestCo)	Visible	Visible	Visible	Visible
Cert B (Active, OtherCo)	Visible	**NOT visible**	Visible	Visible
Cert C (Draft, TestCo)	**NOT visible**	**NOT visible**	Visible	Visible
Cert D (Archived, TestCo)	**NOT visible**	**NOT visible**	Visible	Visible
Internal tags shown	No	No	Yes	Yes
External tags shown	Yes	Yes	Yes	Yes
Edit button	No	No	Yes	Yes
Delete button	No	No	No	Yes
Filter by internal tag	No	No	Yes	Yes
Filter by all companies	Yes	No (own only)	Yes	Yes
Filter by all statuses	No (Active only)	No (Active only)	Yes (all 14)	Yes (all 14)

PART B2: ACCESSIBILITY TESTING

Module 20: Accessibility Testing (WCAG 2.1 AA)

Standard: WCAG 2.1 Level AA | **Tools**: Screen reader, keyboard, browser zoom, contrast analyzer

20.1 Keyboard Navigation

ID	Description	Steps	Expected Result	Status
TC-A11Y-001	Tab order - Login page	1. Navigate to /Account/Login 2. Press Tab repeatedly	Focus: Username → Password → Remember Me → Sign In → Forgot Password	
TC-A11Y-002	Tab order - Public tile view	1. Navigate to public tile view 2. Tab	Focus: Skip link → Search → Filters → Tiles → Load more	
TC-A11Y-003	Tab order - Admin pages	1. Navigate to any Admin page 2. Tab	Focus: Nav → Action buttons → Search → Filters → Table → Pagination	
TC-A11Y-004	Skip navigation link	1. Press Tab on any page	First focusable element is "Skip to main content" link	
TC-A11Y-005	Enter/Space activates buttons	1. Tab to button 2. Press Enter or Space	Button activates	

ID	Description	Steps	Expected Result	Status
TC-A11Y-006	Escape closes modals	1. Open any modal 2. Press Escape	Modal closes, focus returns to trigger	
TC-A11Y-007	Arrow keys in dropdowns	1. Tab to dropdown 2. Use Up/Down arrows	Options navigate via arrow keys	
TC-A11Y-008	Focus trap in modals	1. Open modal 2. Tab through all elements	Focus stays within modal	

20.2 Screen Reader Compatibility

ID	Description	Steps	Expected Result	Status
TC-A11Y-009	Page title announced	1. Navigate to each page with screen reader	Screen reader announces page title	
TC-A11Y-010	Heading hierarchy	1. Use screen reader heading list	Single H1, logical H2/H3, no skipped levels	
TC-A11Y-011	Form labels on login	1. Tab to username/password with screen reader	Fields announced with labels	
TC-A11Y-012	Form labels on certificate create	1. Tab through Create form with screen reader	All fields have associated labels	
TC-A11Y-013	Table headers announced	1. Navigate admin table with screen reader	Column headers announced	
TC-A11Y-014	Button purposes announced	1. Focus on action buttons	Screen reader announces button purpose	
TC-A11Y-015	Status badge alt text	1. Navigate to certificate tile	Status badge text announced	
TC-A11Y-016	Image alt text	1. Navigate to certificate detail with thumbnail	Image has alt text	
TC-A11Y-017	sr-only labels for search/filter	1. Focus on search/filter inputs	Screen reader labels are read	

20.3 Color Contrast & Visual

ID	Description	Steps	Expected Result	Status
TC-A11Y-018	Text contrast ratio	1. Use contrast checker on body text	Contrast >= 4.5:1 for normal text, >= 3:1 for large text	
TC-A11Y-019	Button contrast	1. Check primary button text vs background	Contrast >= 4.5:1	
TC-A11Y-020	Status badge contrast	1. Check Active, Draft, Expired badge combos	All meet >= 4.5:1 contrast	
TC-A11Y-021	Link visibility	1. Inspect links	Links distinguishable by more than color alone	
TC-A11Y-022	Focus indicator visible	1. Tab through elements	Visible focus ring on every focusable element	
TC-A11Y-023	Information not by color alone	1. Check validation errors, status indicators	Errors have icon + text. Statuses have label text.	

ID	Description	Steps	Expected Result	Status
TC-A11Y-024	16px minimum font	1. Inspect all text	No text smaller than 16px	

20.4 ARIA Attributes

ID	Description	Steps	Expected Result	Status
TC-A11Y-025	aria-label on search inputs	1. Inspect search inputs	Each search has aria-label	
TC-A11Y-026	aria-label on filter selects	1. Inspect filter dropdowns	Each filter has aria-label	
TC-A11Y-027	aria-expanded on collapsible panels	1. Inspect filter panel toggle	aria-expanded="true"/"false"	
TC-A11Y-028	aria-hidden on decorative icons	1. Inspect Font Awesome icons	Decorative icons have aria-hidden="true"	
TC-A11Y-029	role="alert" on notifications	1. Trigger toast 2. Inspect	Has role="alert" or aria-live="polite"	
TC-A11Y-030	role="dialog" on modals	1. Open modal 2. Inspect	Has role="dialog" and aria-labelledby	
TC-A11Y-031	aria-invalid on validation errors	1. Submit form with errors 2. Inspect	Field has aria-invalid="true"	

20.5 Form Accessibility

ID	Description	Steps	Expected Result	Status
TC-A11Y-032	Required fields indicated	1. View cert create form	Required fields have visual * AND aria-required="true"	
TC-A11Y-033	Error messages linked to fields	1. Submit with validation errors	Each error associated with field via aria-describedby	
TC-A11Y-034	Autocomplete accessible	1. Type in search 2. Use arrow keys	Suggestions navigable via keyboard. aria-autocomplete present.	
TC-A11Y-035	File upload accessible	1. Tab to upload zone	Focusable, activatable via Enter/Space	
TC-A11Y-036	Date picker accessible	1. Tab to date fields 2. Enter date via keyboard	Date enterable via keyboard	
TC-A11Y-037	Tag selection accessible	1. Tab to tag selector 2. Use keyboard	Tags selectable/removable via keyboard	
TC-A11Y-038	Checkbox filters accessible	1. Tab to checkboxes 2. Press Space	Checkboxes toggle via Space	

20.6 Modal & Dialog Accessibility

ID	Description	Steps	Expected Result	Status
TC-A11Y-039	Delete confirmation modal	1. Click Delete 2. Inspect	Title, text, labeled buttons. Focus trapped.	
TC-A11Y-040	Help modal accessible	1. Open Help modal	Content readable, close accessible, Escape works	

ID	Description	Steps	Expected Result	Status
TC-A11Y-041	Modal focus management	1. Open any modal	Focus moves to modal. On close, returns to trigger.	
TC-A11Y-042	Toast notification announced	1. Trigger success action	Toast announced via aria-live region	

20.7 Responsive & Zoom Accessibility

ID	Description	Steps	Expected Result	Status
TC-A11Y-043	200% zoom usable	1. Zoom to 200% 2. Navigate tile view	Content reflows, no horizontal scroll. All functions work.	
TC-A11Y-044	200% zoom admin tables	1. Zoom to 200% on admin page	Table scrollable or reformatted. No content cut off.	
TC-A11Y-045	Touch target size (mobile)	1. View on mobile/responsive	All interactive elements >= 44x44px	
TC-A11Y-046	Text resizing	1. Use browser text-only zoom to 200%	Text resizes without breaking layout	

20.8 Data Table Accessibility

ID	Description	Steps	Expected Result	Status
TC-A11Y-047	Admin table has <th> headers	1. Inspect admin table HTML	Proper <thead>/<th> elements	
TC-A11Y-048	Table has caption or aria-label	1. Inspect table element	Table has <caption> or aria-label	
TC-A11Y-049	Sortable columns announced	1. If sortable, tab to header	Screen reader announces sort state	
TC-A11Y-050	Empty table state	1. Apply filter with no results	Accessible "No results found" message	

PART C: TRACEABILITY & REPORTING

24. Requirements Traceability Matrix

Core Requirements Mapping

Requirement	Description	Test Cases	Verified?
REQ-AUTH-01	Role-based access control (6 roles)	TC-PERM-001 to TC-PERM-046	
REQ-AUTH-02	Two-factor authentication	TC-AUTH-009 to TC-AUTH-012	
REQ-AUTH-03	Session management & timeout	TC-AUTH-015, TC-AUTH-018	
REQ-CERT-01	Certificate CRUD operations	TC-ANL-010 to TC-ANL-047	
REQ-CERT-02	14-state workflow	TC-WF-001 to TC-WF-047	
REQ-CERT-03	Certificate number format (YY-NNNN)	TC-ANL-011, TC-NEG-019, TC-PUB-017	
REQ-CERT-04	PDF upload with preview	TC-ANL-021 to TC-ANL-023, TC-ANL-049	
REQ-CERT-05	Product identifier management	TC-ANL-026 to TC-ANL-030	
REQ-CERT-06	Tag management (internal/external)	TC-ANL-024/025, TC-ADM-043 to 051	

Requirement	Description	Test Cases	Verified?
REQ-CERT-07	Custom fields per cert type	TC-ANL-017, TC-ADM-035 to 042	
REQ-CERT-08	Certification Results field (optional)	TC-ANL-031	
REQ-CERT-09	Certificate number displayed (not ID)	TC-ANL-052, TC-PUB-017	
REQ-VIEW-01	Public tile view with infinite scroll	TC-PUB-001 to TC-PUB-012	
REQ-VIEW-02	External tile view (company-scoped)	TC-EXT-001 to TC-EXT-012	
REQ-VIEW-03	Internal tile view (all data)	TC-ANL-001 to TC-ANL-009	
REQ-VIEW-04	Autocomplete search	TC-PUB-003, TC-PUB-004	
REQ-VIEW-05	Shareable URLs	TC-PUB-009, TC-ANL-008	
REQ-VIEW-06	Database-driven filters	TC-PUB-005 to TC-PUB-008	
REQ-EXT-01	External sees only own company	TC-EXT-002, TC-EXT-003, TC-EXT-017	
REQ-EXT-02	External cannot see internal tags	TC-EXT-007, TC-PUB-011	
REQ-EXT-03	External no Add/Edit/Delete	TC-EXT-004 to 006, TC-EXT-018/019	
REQ-ADM-01	User management (CRUD + import)	TC-ADM-004 to TC-ADM-018	
REQ-ADM-02	Role management with permissions	TC-ADM-019 to TC-ADM-027	
REQ-ADM-03	Certificate type hierarchy	TC-ADM-028 to TC-ADM-034	
REQ-ADM-04	Partner management (core entities)	TC-ADM-052 to TC-ADM-058	
REQ-ADM-05	Partners - export/add only (no import)	TC-ADM-058, TC-IMP-007/008	
REQ-ADM-06	Admin role cannot be deleted	TC-ADM-024	
REQ-ADM-07	Unique emails per user	TC-ADM-010	
REQ-EXP-01	CSV export only (no Excel/JSON/PDF)	TC-EXP-008	
REQ-IMP-01	CSV import with templates	TC-IMP-001 to TC-IMP-006	
REQ-API-01	Partner API with key authentication	TC-API-001 to TC-API-004	
REQ-API-02	Certificate REST endpoints	TC-API-005 to TC-API-014	
REQ-API-03	Product ID search via API	TC-API-012	
REQ-API-04	Rate limiting	TC-API-020, TC-API-021	
REQ-UI-01	No native alert/confirm/prompt	TC-NEG-011 to TC-NEG-017	
REQ-UI-02	Delete: confirm modal + toast	TC-NEG-011 to TC-NEG-014	
REQ-UI-03	16px minimum font size	TC-NEG-024	
REQ-SEC-01	XSS prevention	TC-NEG-021	
REQ-SEC-02	SQL injection prevention	TC-NEG-022	
REQ-A11Y-01	Keyboard navigation	TC-A11Y-001 to TC-A11Y-008	
REQ-A11Y-02	Screen reader compatibility	TC-A11Y-009 to TC-A11Y-017	
REQ-A11Y-03	Color contrast >= 4.5:1	TC-A11Y-018 to TC-A11Y-024	
REQ-ADM-08	Certificate Import Admin-only	TC-ANL-059, TC-PERM-019, TC-ADM-075 to 081	

Requirement	Description	Test Cases	Verified?
REQ-ADM-09	Delete with dependency protection	TC-ADM-085 to TC-ADM-089	
REQ-AMZN-01	Amazon Code in API responses	TC-API-016a, TC-API-016c, TC-API-016e	Built & Deployed
REQ-AMZN-02	Pathway field (admin-managed)	TC-ADM-065 to TC-ADM-068c	Built & Deployed
REQ-AMZN-03	Pathway on certificate form	TC-ANL-031c to TC-ANL-031e	Built & Deployed
REQ-AMZN-04	Pathway in API responses	TC-API-016b, TC-API-016d, TC-API-016f	Built & Deployed
REQ-AMZN-05	Soft validation for Amazon cert results	TC-ANL-031f	Pending Q4
REQ-AUDIT-01	Field-level change tracking	TC-AUDIT-001	Covered
REQ-AUDIT-02	Certificate status transition history	TC-AUDIT-002	Covered
REQ-AUDIT-03	User identification on all changes	TC-AUDIT-003	Covered
REQ-AUDIT-07	Status history on cert detail page	TC-AUDIT-007	Gap - UI
REQ-AUDIT-08	Admin audit log viewer	TC-AUDIT-008	Gap - UI

End-to-End Scenario Coverage

Requirement	E2E Scenario	Steps
Full certificate lifecycle	E2E Scenario 1 (Module 11)	E2E-LC-001 to E2E-LC-046
Company onboarding & data isolation	E2E Scenario 2 (Module 12)	E2E-CO-001 to E2E-CO-026
Partner API integration	E2E Scenario 3 (Module 13)	E2E-API-001 to E2E-API-018
Admin system setup feeds into cert creation	E2E Scenario 4 (Module 14)	E2E-SETUP-001 to E2E-SETUP-027
Automated renewal & expiration	E2E Scenario 5 (Module 15)	E2E-REN-001 to E2E-REN-014
Rejection, correction, re-approval	E2E Scenario 6 (Module 16)	E2E-REJ-001 to E2E-REJ-014
Bulk template → import → verify → export	E2E Scenario 7 (Module 17)	E2E-BULK-001 to E2E-BULK-024
User account lifecycle	E2E Scenario 8 (Module 18)	E2E-USER-001 to E2E-USER-026
Same data viewed across all 4 roles	E2E Scenario 9 (Module 19)	E2E-SEARCH-001 to E2E-SEARCH-026

25. Defect Reporting Template

When a test fails, log a defect using this template:

```
Defect Report

Defect ID:    DEF-XXX
Date:         YYYY-MM-DD
Tester:       [Name]
Test Case ID: TC-XXX-XXX
Severity:     Critical / High / Medium / Low
Priority:     P1 / P2 / P3 / P4

Summary:
[One-line description of the defect]

Environment:
- Browser: [Chrome/Firefox/Edge + version]
- URL: [Environment URL]
- Role: [Admin/Analyst/External/Public]

Steps to Reproduce:
1. [Step 1]
2. [Step 2]
3. [Step 3]

Expected Result:
[What should have happened]

Actual Result:
[What actually happened]

Screenshots/Evidence:
[Attach screenshots or console errors]

Additional Notes:
[Any context, workarounds, or related defects]
```

26. Test Execution Summary

Part A: Feature-Level Testing

Module	Total Tests	Pass	Fail	Blocked	Not Run	% Complete
1. Authentication	18				18	0%
2. Public User	20				20	0%
3. External User	24				24	0%
4. Analyst User	69				69	0%
5. Admin User	95				95	0%
6. Workflow (14 States)	47				47	0%
7. Partner API	27				27	0%
8. Permissions Matrix	47				47	0%
9. Export/Import	18				18	0%
10. Edge Cases & Negative	26				26	0%

Module	Total Tests	Pass	Fail	Blocked	Not Run	% Complete
10b. Audit Trail	9				9	0%
Part A Subtotal	**400**				**400**	**0%**

Part B: End-to-End Business Scenarios

Module	Total Steps	Pass	Fail	Blocked	Not Run	% Complete
11. Certificate Full Lifecycle	46				46	0%
12. New Company Onboarding	26				26	0%
13. Partner API Integration	18				18	0%
14. Admin System Setup	27				27	0%
15. Certificate Renewal	14				14	0%
16. Rejection & Recovery	14				14	0%
17. Bulk Operations	24				24	0%
18. User Access Lifecycle	26				26	0%
19. Search & Discovery Cross-Role	26				26	0%
Part B Subtotal	**221**				**221**	**0%**

Part B2: Accessibility Testing

Module	Total Tests	Pass	Fail	Blocked	Not Run	% Complete
20. Accessibility (WCAG 2.1 AA)	50				50	0%
Part B2 Subtotal	**50**				**50**	**0%**

Combined Total

Category	Total	Pass	Fail	Blocked	Not Run	% Complete
Part A: Feature Tests	400				400	0%
Part B: E2E Scenarios	221				221	0%
Part B2: Accessibility	50				50	0%
GRAND TOTAL	**671**				**671**	**0%**

Sign-Off

Role	Name	Date	Signature
Client Representative			

This guide covers the complete GCC Database Solution application. All test cases should be executed per role before UAT sign-off.